

Fruits and Seeds

The Fruit

A fruit is a post-fertilization modification of the ovary; if formed without fertilization it is **parthenocarpic**. The fruit wall is the **pericarp** (epicarp, mesocarp, endocarp).

Example: Drupe

Mango and coconut are **drupes** from monocarpellary superior ovaries with a single seed; mango has fleshy mesocarp, coconut has fibrous mesocarp.

Dicot Seeds

Have a seed coat (testa + tegmen) and embryo with two **cotyledons**; often non-endospermous (bean, gram, pea).

Monocot Seeds

Have a dominant endosperm with aleurone layer, a single shield-shaped cotyledon (**scutellum**), and protective sheaths — **coleoptile** (plumule) and **coleorrhiza** (radicle).

True Fruits vs False Fruits

A **true fruit** develops solely from the ovary (e.g., mango), while a **false fruit** develops from the ovary along with other floral parts like the thalamus (e.g., apple, where the fleshy part comes from the receptacle).

Fruit Classification Overview

Fruits are broadly classified as **simple fruits** (from a single ovary, e.g., mango, guava), **aggregate fruits** (from multiple carpels of one flower, e.g., custard apple), and **composite fruits** (from an entire inflorescence, e.g., pineapple, fig).

Dicot vs Monocot Seed

Feature	Dicot Seed	Monocot Seed
Cotyledons	Two	One (scutellum)
Endosperm	Often absent	Usually present
Protective sheaths	Absent	Coleoptile & coleorrhiza

3 Layers of Pericarp

- Epicarp (outer)
- Mesocarp (middle)
- Endocarp (inner)